

Deep Generative Models

Lecture 3



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Deep Generative Models – Lecture 3



■ Outline

1. Deepfakes

- Production techniques
- Detection methods



What is a deepfake?



- **Definition:**

“Manipulated or synthetic audio or visual media that seem authentic, and which feature people that appear to say or do something they have never said or done, produced using artificial intelligence techniques, including machine learning and deep learning.”

[\[European Parliament Research Service, 2021\]](#)

- Multiple definitions exist in literature (+ many legal definitions)
- ...but, still ambiguous, boundary between real/fake content may be fuzzy
- Malicious intent
 - Disinformation campaigns, fake news
 - Various frauds
 - Fake porn
- Potentially harmful for both individuals and society



Deepfake Production Techniques

Deepfake Production Techniques



- Images
 - Advanced image editing tools
 - Entire image synthesis (text2image, image2image)

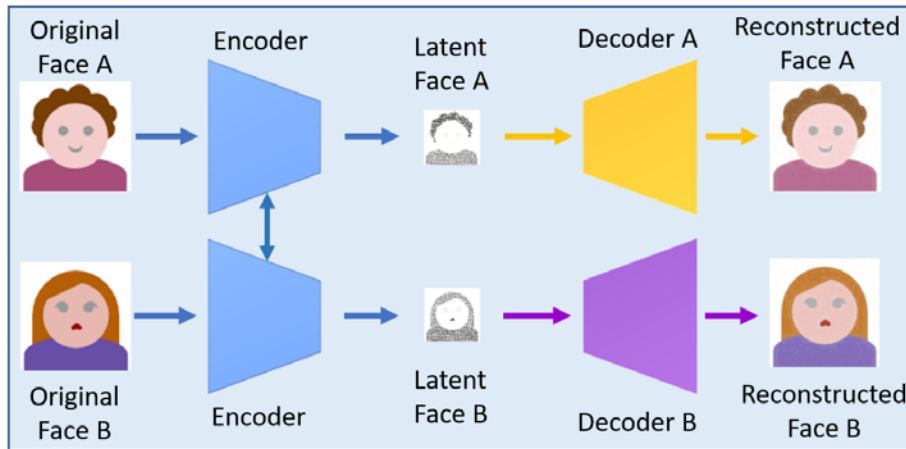
- Video (people, faces)
 - “Classical” deepfakes (historical approach)
 - Face swap
 - Face reenactment
 - LipSync
 - Entire video synthesis (text2video, image2video, video2video)



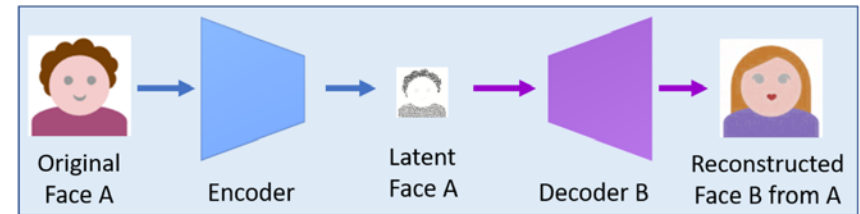
“Classical” Deepfake

- Seamless swapping a face in an image/video, e.g. [[Nguyen et al., 2020](#)]
- Auto-encoder architecture
 - Single shared encoder (to capture pose / expressions)
 - Two decoders (Source and Target to capture person’s identity)

Training



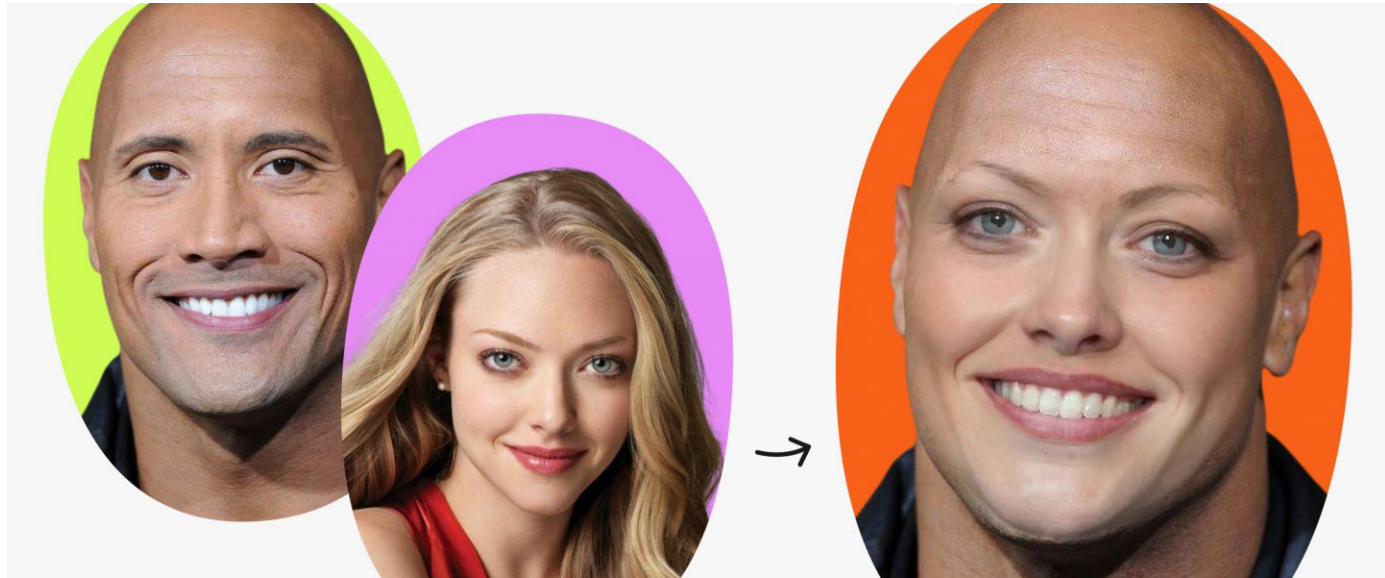
Deployment





Face Swap

- Single/multiple images (identity) → target image/video (pose, expression)
 - Typically identity swap in latent space
 - Only inner face changed (hair, head shape remain the same)
 - Blending always present

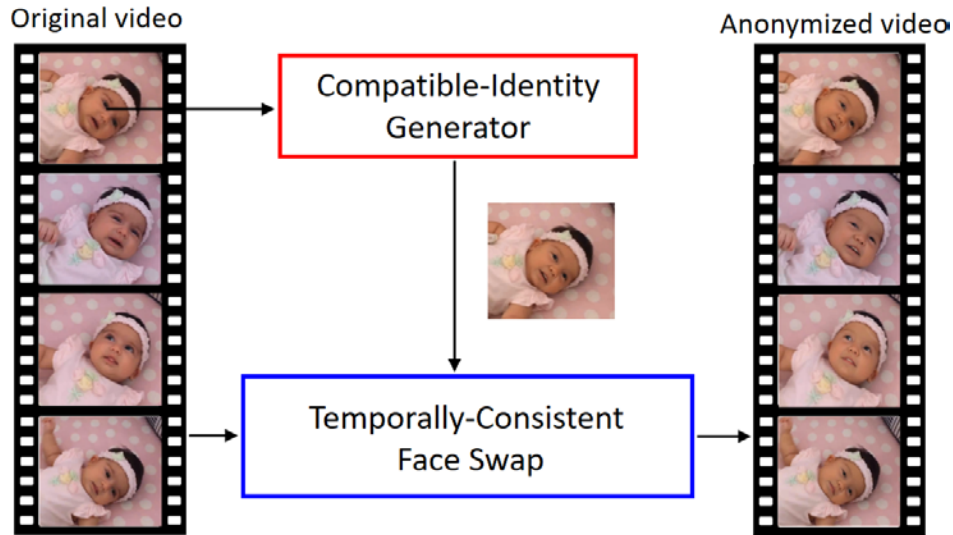


Face Swapper

Face Swap – Anonymization (De-identification)



- BLANKET: Anonymizing Faces in Infant Video Recordings [[Hadera-2025](#)]



1. Stable Diffusion Inpainting
2. Face Swap by [FaceFusion](#)





Face Reenactment

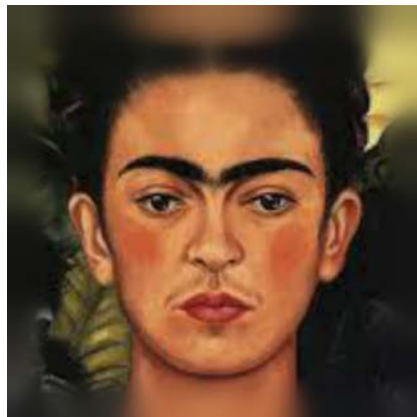
- Driving video (“motion”) + image (“appearance”) → video
 - Warps the input image by the driving video



[\[video\]](#)



[Liveportraits](#)



[\[video\]](#)



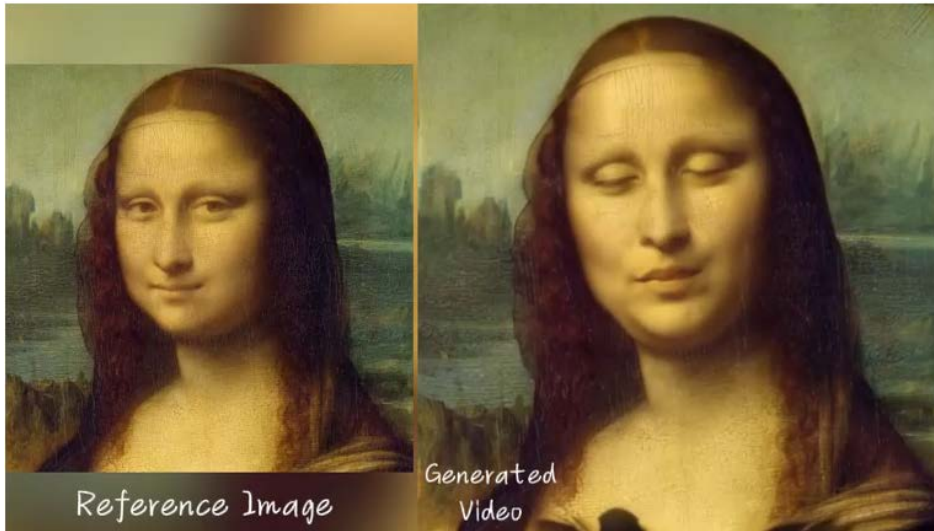
[\[Jahoda-2024\]](#) (uses StyleGAN2)



LipSync

- Input video + different audio → LipSynced video
- Model synthesises realistic mouth shapes (visemes) aligned with speech
- Could be used for automatic movie dubbing (different language versions)
- Sometimes head motion is synthesized too (e.g. talking heads)
- Synthesized speech (voice cloning) often used
- Many tools available, e.g. Wav2lip

[[video](#)]



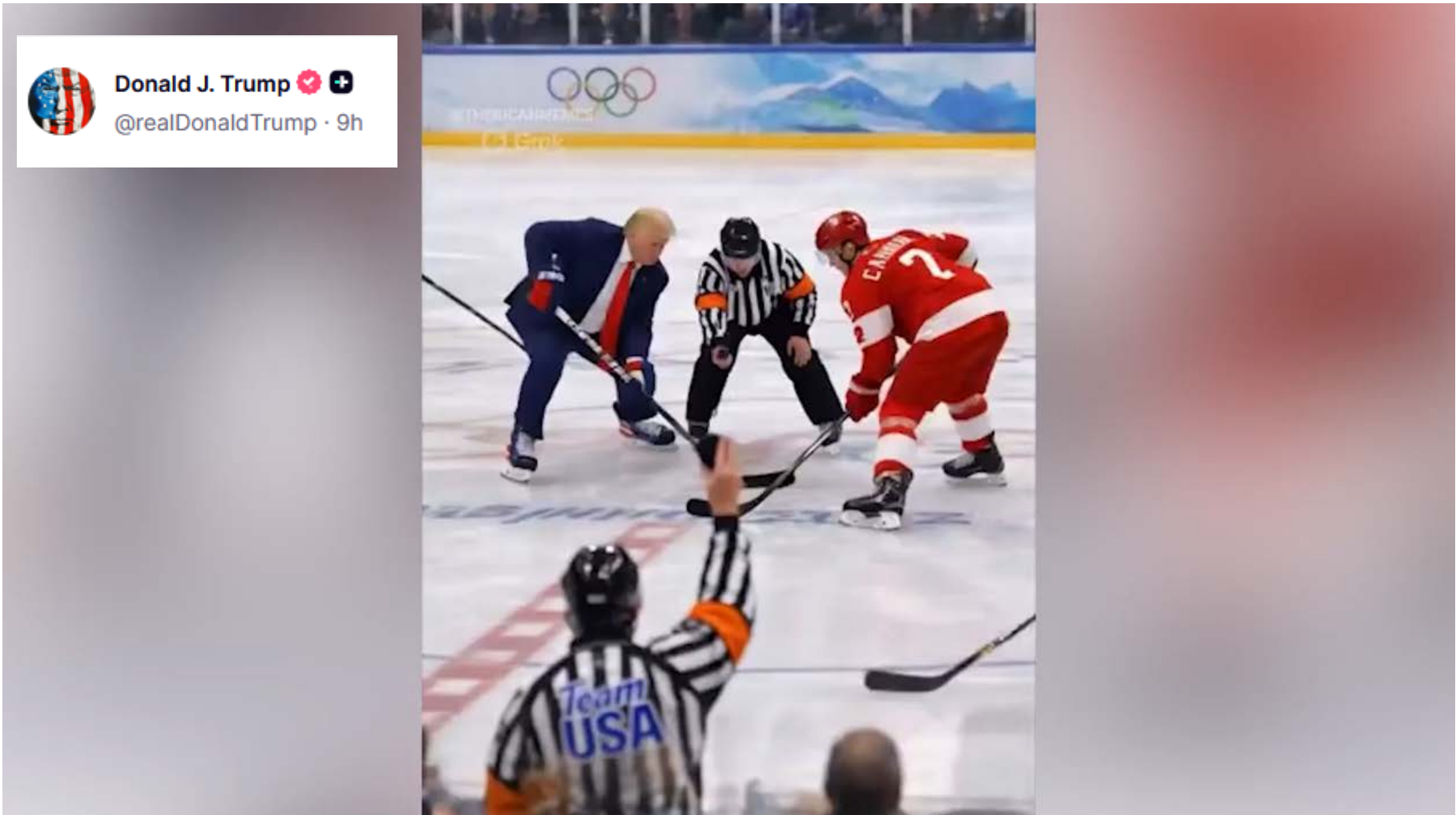
[[video](#)] [[Tian-2024](#)]





Entire Video Synthesis

- Entire video synthesis has progressed a lot recently
 - Many tools (commercial and open-source)...



[video]



Deepfake Detection



Sometimes the artifacts are obvious...

- Detecting and Correcting Perceptual Artifacts [[Šubrtová-2024](#)]



Sometimes the artifacts are obvious...

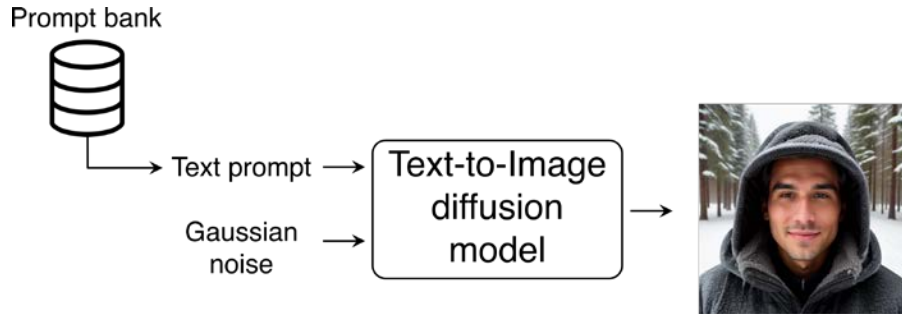
- Detecting and Correcting Perceptual Artifacts [[Šubrtová-2024](#)]



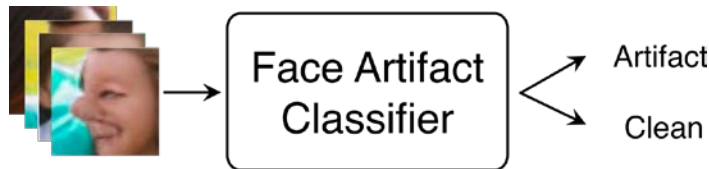


Detecting and Correcting Perceptual Artifacts

- Synthetic dataset of automatically generated scene with people

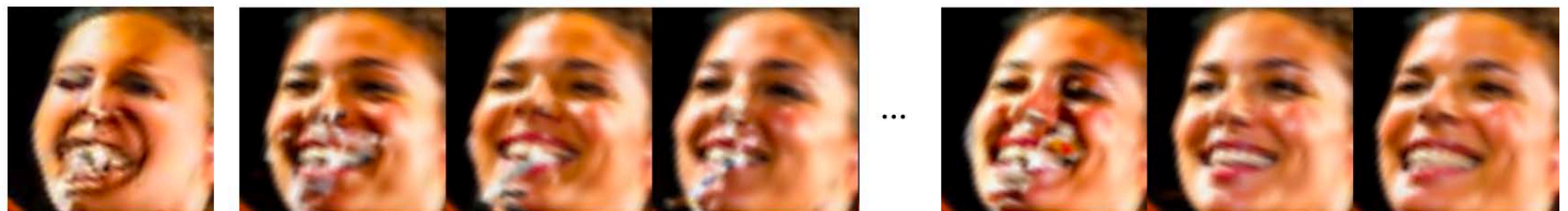


- Face classifier (Artifact vs. clean)



Model	Acc	AUC
Face-size	0.8731	0.9213
Laion Aesthetics [25]	0.8134	0.9420
Face detector score	0.5896	0.6475
PAL4VST [33] (face crops)	0.7164	0.7981
Synth/Real (last layer finetuned)	0.7761	0.8651
Ours	0.9254	0.9678

- Correction by SD inpainting (zoom, multiple attempts, classifier response)



Original

more artifacts

fewer artifacts



Detecting Synthetic Face Images

- Detection of Synthetic Face Images: Accuracy, Robustness, Generalization [[Petrzelkova-2025](#)]
- Experimental setup
 - Dataset of synthetic face images
 - 5 generators:
 - 2 Diffusion models (Stable Diffusion, Realistic vision)
 - 2 GANs (Progressive GAN, StyleGAN)
 - 1 commercial app (FaceApp)
 - Real images FFQ-dataset
 - 2k images each generator (and real set)
 - Resolution 224 x 224 pixels



RV5.1



SD1.4



FA



PG



SG





Detector Generalization

- Simple ResNET-50 detector trained (fake/real)
- In-dataset / Cross-dataset setting

		Test set				
		RV5.1	SD1.4	FA	PG	SG
Training set	RV5.1	100	58	49	50	50
	SD1.4	51	100	50	54	49
	FA	53	50	80	87	60
	PG	49	61	54	100	50
	SG	48	48	54	66	94

(a) Training on a single generator

		Test set				
		RV5.1	SD1.4	FA	PG	SG
Training set	-RV5.1	58	92	80	94	91
	-SD1.4	91	84	85	91	91
	-FA	95	94	55	94	89
	-PG	93	92	77	79	85
	-SG	95	95	80	94	52

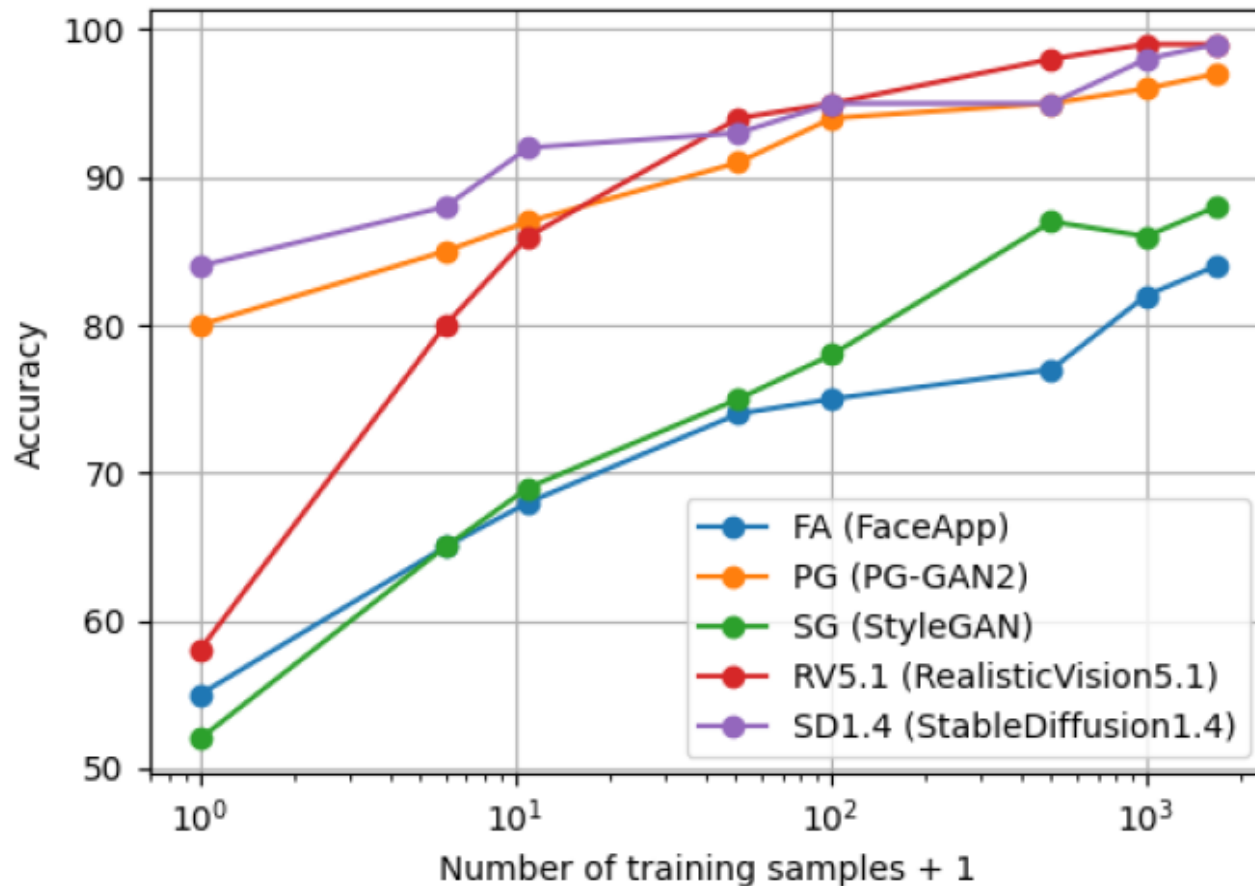
(b) Leave one out training

- ⇒ Perfect accuracy for known generator (generator leaves invisible “unique” fingerprint)
- ⇒ Limited generalization to unseen generator



Detector Generalization

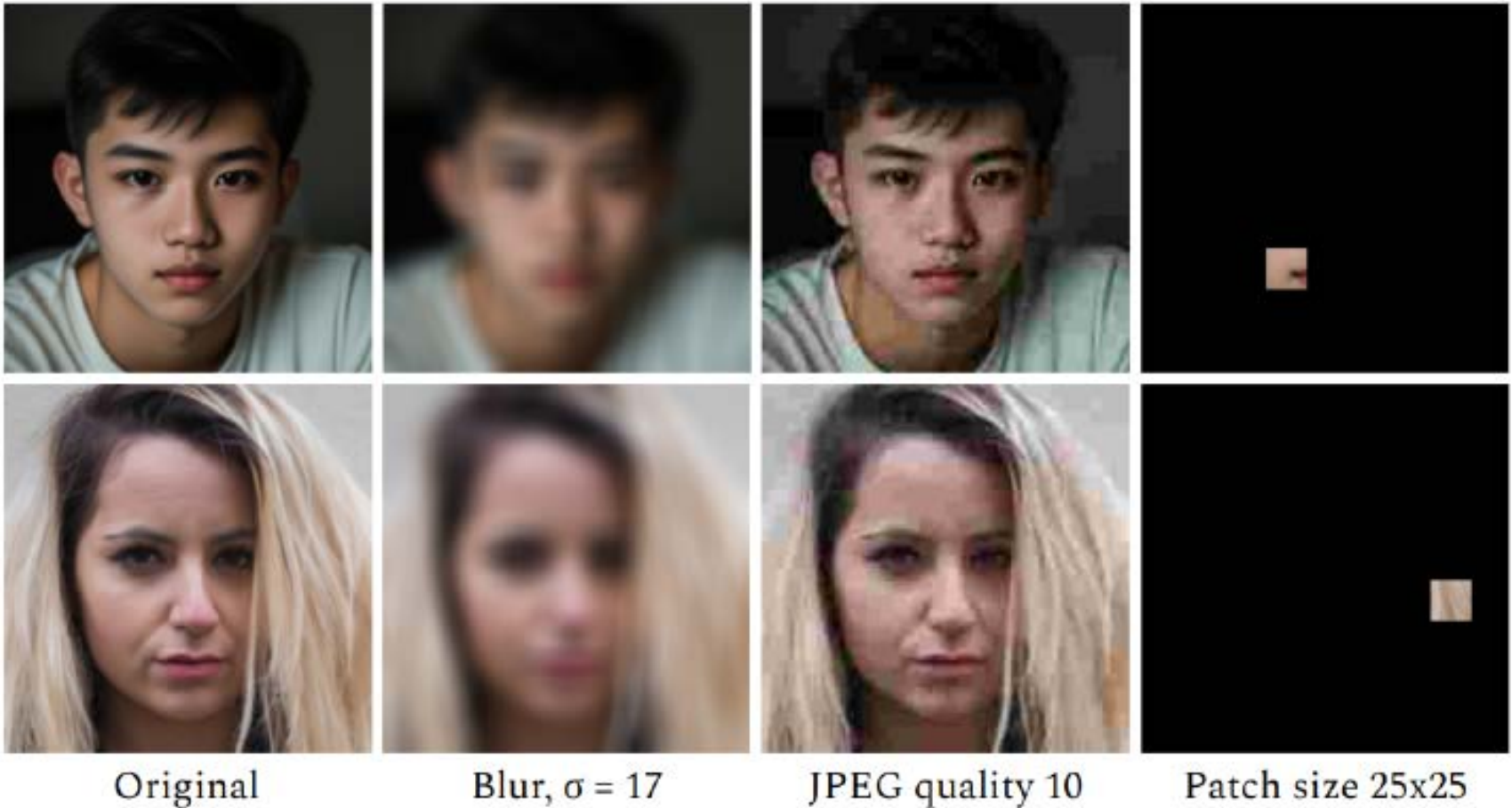
- Learning curve is steep
 - Only few images of an unseen generator is enough to recover detection accuracy





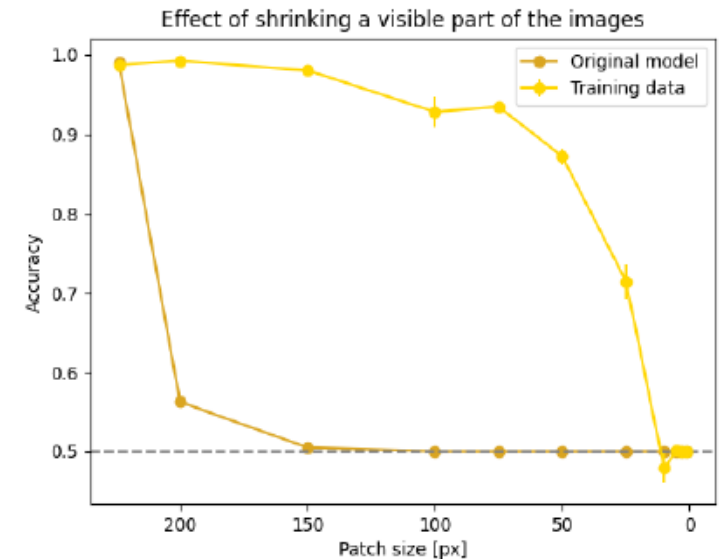
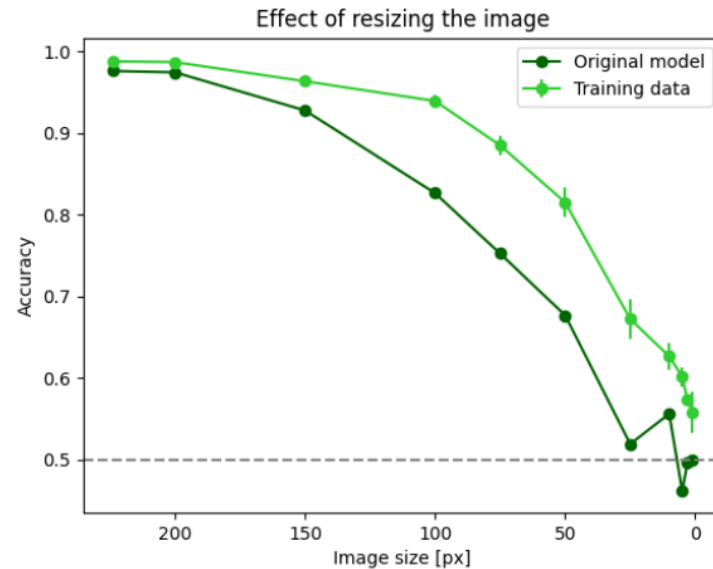
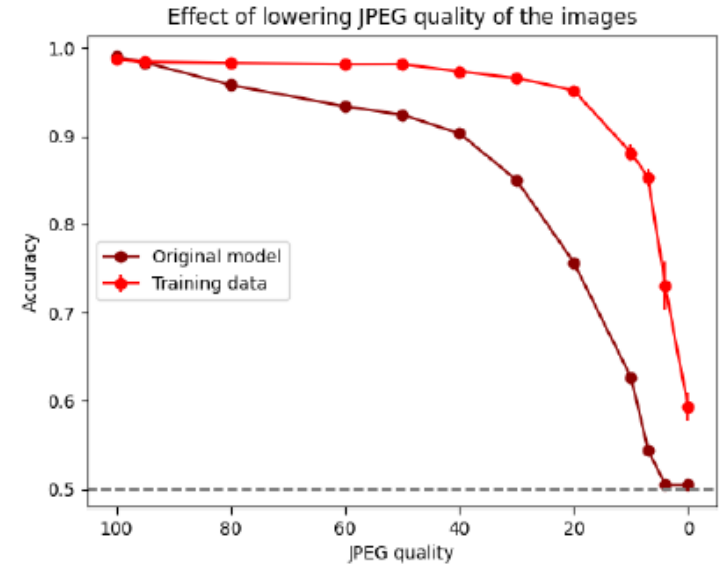
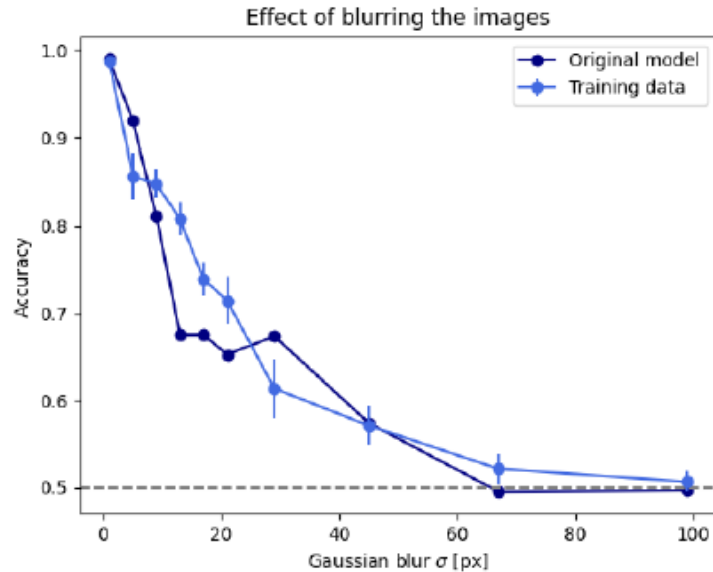
Detector Robustness

- “Laundering attacks” (to erase the generator fingerprint)





Detector Robustness

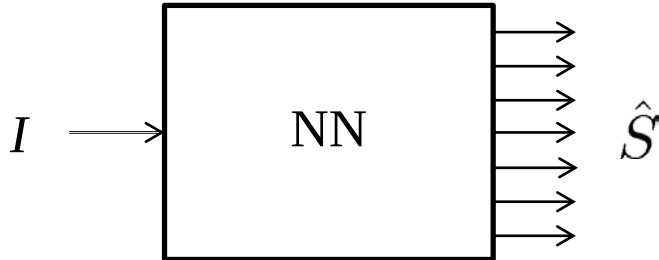


=> Surprisingly high robustness (for seen generators)

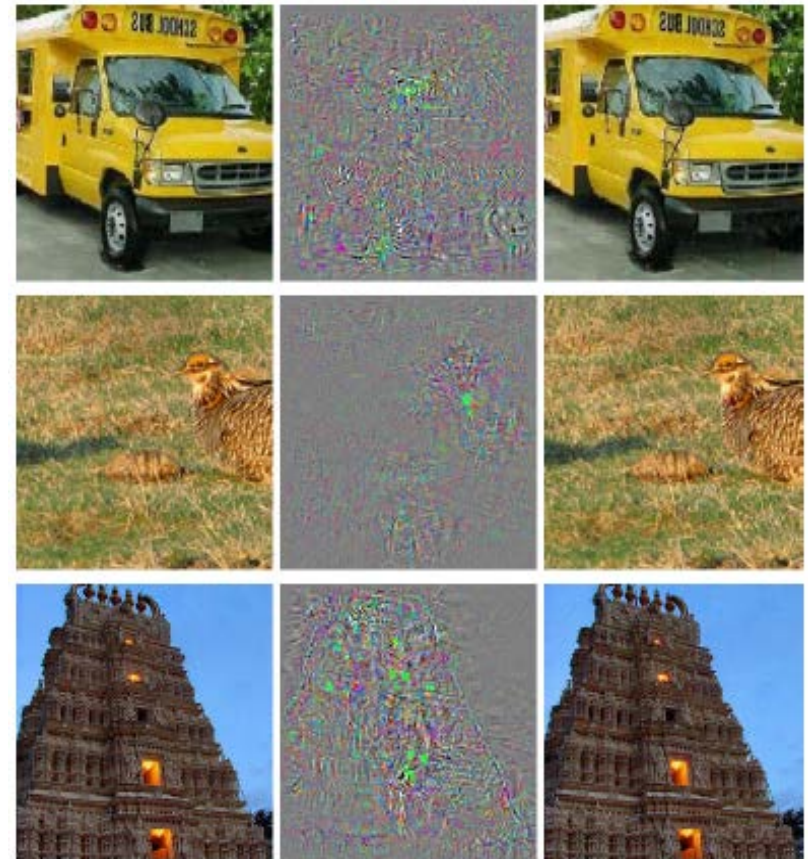


Detector Robustness

- Adversarial attacks
 - Any deep network can easily be fooled [[Szegedy-ICLR-2014](#)]
 - Small perturbation of the input image changes the output of the trained “well-performing”
 - The perturbation is a non-random image, imperceptible for human



$$\min_r \{ ||\text{NN}(I + r) - S||^2 + \lambda ||r||^2 \}$$



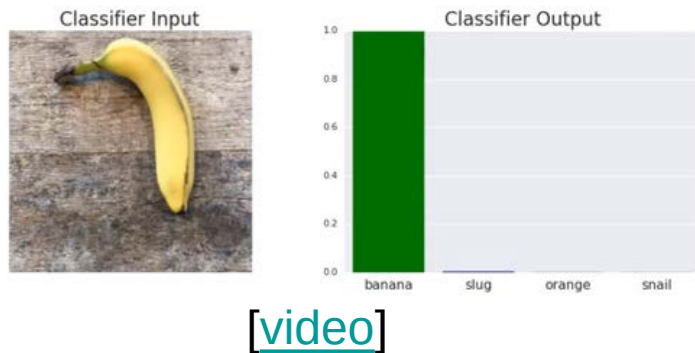
I r $I + r$



Adversarial attacks (cont.)

- Adversarial physical attacks

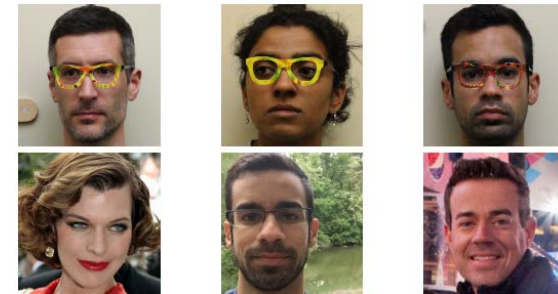
Adversarial sticker [Brown-2018]



Adversarial T-shirt [Xu-2019]



Adversarial glasses [Sharif-2016]



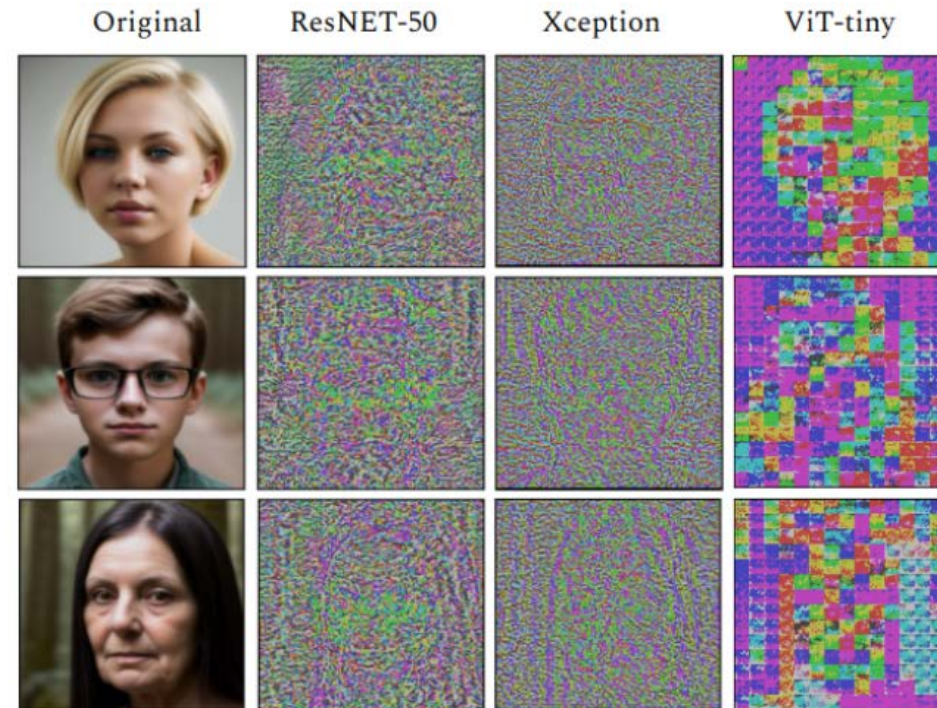


Detector Robustness

- Adversarial noise found for 3 different detectors (fake→real)

confusion rate

FGSM $\epsilon = 0.05$	ResNET-50	Xception	ViT-tiny
ResNET-50	100.00	100.0	15.69
Xception	100.00	100.0	10.22
ViT-tiny	100.00	100.0	100.0



=> Sensitive to adversarial attacks

=> Cross-architecture (black box) attacks work!



Localizing Partial Manipulations

- Seamless mix of real and fake content (Inpainting by SD)

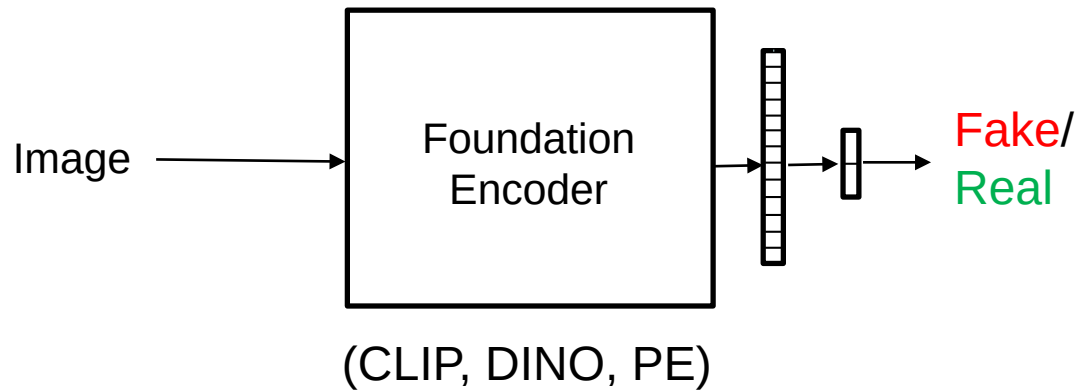


- Simple YOLO-based detector
=> Almost perfect localization accuracy for known generation model

State-of-the-art Deepfake Detectors



- SotA DF detection approach: Learn a classifier on top of a pretrained foundation model
- Better generalization to unseen generators is reported



Deepfake Detection that Generalizes across Benchmarks



■ [Yermakov-2026]

Input

Video frames: $f_1, f_2, \dots, f_T$



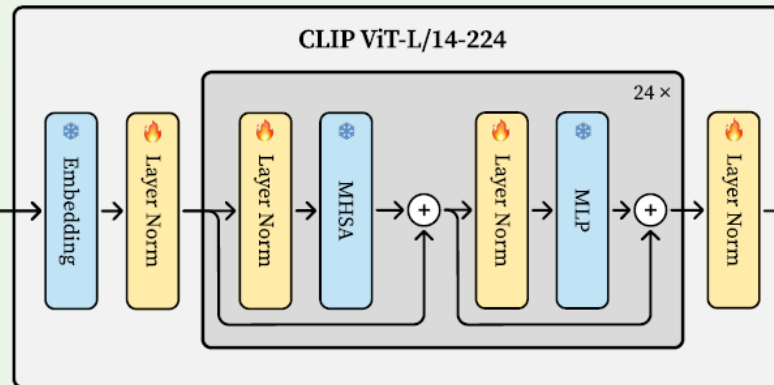
Output

$$\text{Video-level prediction: } p(\text{Fake} | x_1, x_2, \dots, x_T) = \frac{1}{T} \sum_{t=1}^T p(\text{Fake} | x_t)$$

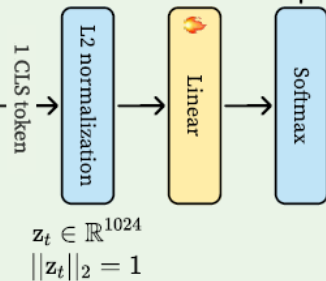
$$\text{Frame-level prediction: } p(\text{Fake} | x_t)$$

Detected and aligned face: x_t

1 CLS token
16x16 patches



1 CLS token



- Layer-norm tuning, only 0.03% of all parameters
- State-of-the-art average cross-dataset accuracy on 14 benchmarks

Deepfake Detection that Generalizes across Benchmarks



■ Cross-dataset video-level AUROC (trained on FF++, 2019)

Model	Year	Publication	Input	Backbone	CDFv2	DFD	DFDC	FFIW
LipForensics [20]	2021	CVPR	Video	ResNet-18	82.4	–	73.5	–
FTCN [66]	2021	ICCV	Video	3D ResNet-50	86.9	–	74.0	74.5 ^[45]
RealForensics [21]	2022	CVPR	Video	Modified CSN	86.9	–	75.9	–
SBI [45]	2022	CVPR	Frame	EFNB4	93.2	82.7	72.4	84.8
AUNet [64]	2023	CVPR	Video	Xception+ART	92.8	99.2	73.8	81.5
StyleDFD [12]	2024	CVPR	Video	3D ResNet-50	89.0	96.1	–	–
LSDA [57]	2024	CVPR	Frame	EFNB4	91.1	–	77.0	72.4 ^[58]
LAA-Net [38]	2024	CVPR	Frame	EFNB4	95.4	98.4	86.9	–
AltFreezing [53]	2024	CVPR	Video	3D ResNet-50	89.5	98.5	–	–
NACO [63]	2024	ECCV	Video	ViT-B/16	89.5	–	76.7	–
RAE [50]	2024	ECCV	Frame	ViT-B/16	95.5	99.0	80.2	–
TALL++ [55]	2024	IJCV	Video	Swin-B	92.0	–	78.5	–
ProDet [11]	2024	NeurIPS	Frame	EFNB4	92.5	–	77.0	–
UDD [18]	2025	AAAI	Frame	CLIP ViT-B/16	93.1	95.5	81.2	–
P&P [59]	2025	CVPR	Video	CLIP ViT-L/14	94.7	96.5	84.3	92.1
DFD-FCG [22]	2025	CVPR	Video	CLIP ViT-L/14	95.0	–	81.8	–
ForAda [13]	2025	CVPR	Frame	CLIP ViT-L/14	95.7	97.2	87.2	–
Effort [58]	2025	ICML	Frame	CLIP ViT-L/14	95.6	96.5	84.3	92.1
GenD (CLIP)	2025	–	Frame	CLIP ViT-L/14	96.0	97.0	87.1	92.8
GenD (PE)	2025	–	Frame	PE _{core} L	95.8	96.5	81.6	93.3
GenD (DINO)	2025	–	Frame	DINOv3 ViT-L/16	92.2	96.6	84.7	94.5

Method	2019 UADFV	2019 DFD	2019 DFDC	2020 FSH	2020 CDFv2	2021 FFIW	2021 KoDF	2021 FAVC	2022 DFDM	2024 PGF	2024 IDF	2024 DSv1.1	2025 DSv2	2025 CDFv3	Mean
ForAda [13]	99.4	97.2	87.3	82.0	95.7	90.6	88.2	93.1	97.1	86.6	90.8	81.8	72.8	75.6	88.4
Effort [58]	97.4	95.2	85.4	91.2	93.2	92.5	88.1	92.4	98.2	84.9	96.0	82.1	64.4	78.7	88.5
GenD (CLIP)	99.2	96.4	86.4	86.6	94.6	91.5	84.9	96.0	99.6	89.6	97.8	90.1	77.7	85.9	91.2
GenD (PE)	97.7	96.8	82.2	87.6	95.0	93.7	85.1	97.3	98.3	92.3	97.9	87.8	78.6	89.5	91.4
GenD (DINO)	98.6	96.2	85.6	88.8	92.5	92.9	89.7	98.4	99.8	92.4	98.2	86.9	79.4	83.5	91.6

Deepfake Detection that Generalizes across Benchmarks



- Better generalization requires paired dataset:
 - Each fake video and its real source

Paired dataset – for each fake video there exists a real counterpart from which this fake was created



Real

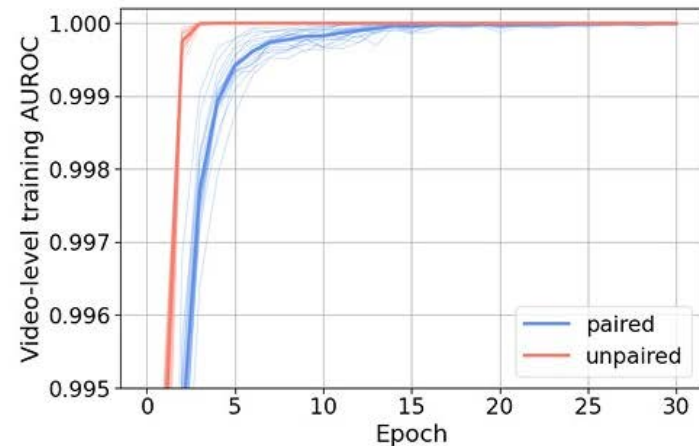
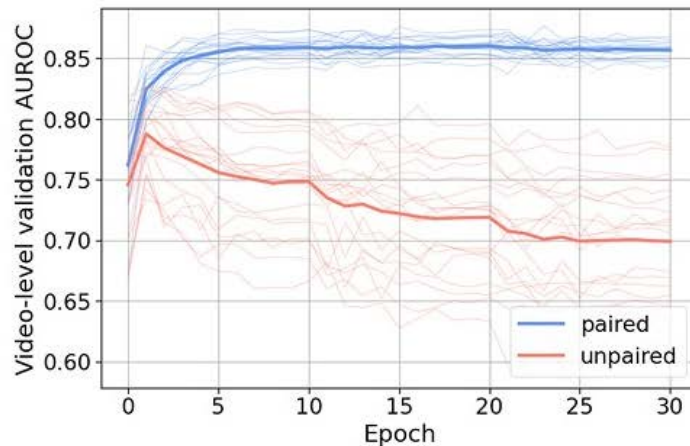
Fake

Unpaired dataset – no real counterpart is present for every fake video



Real

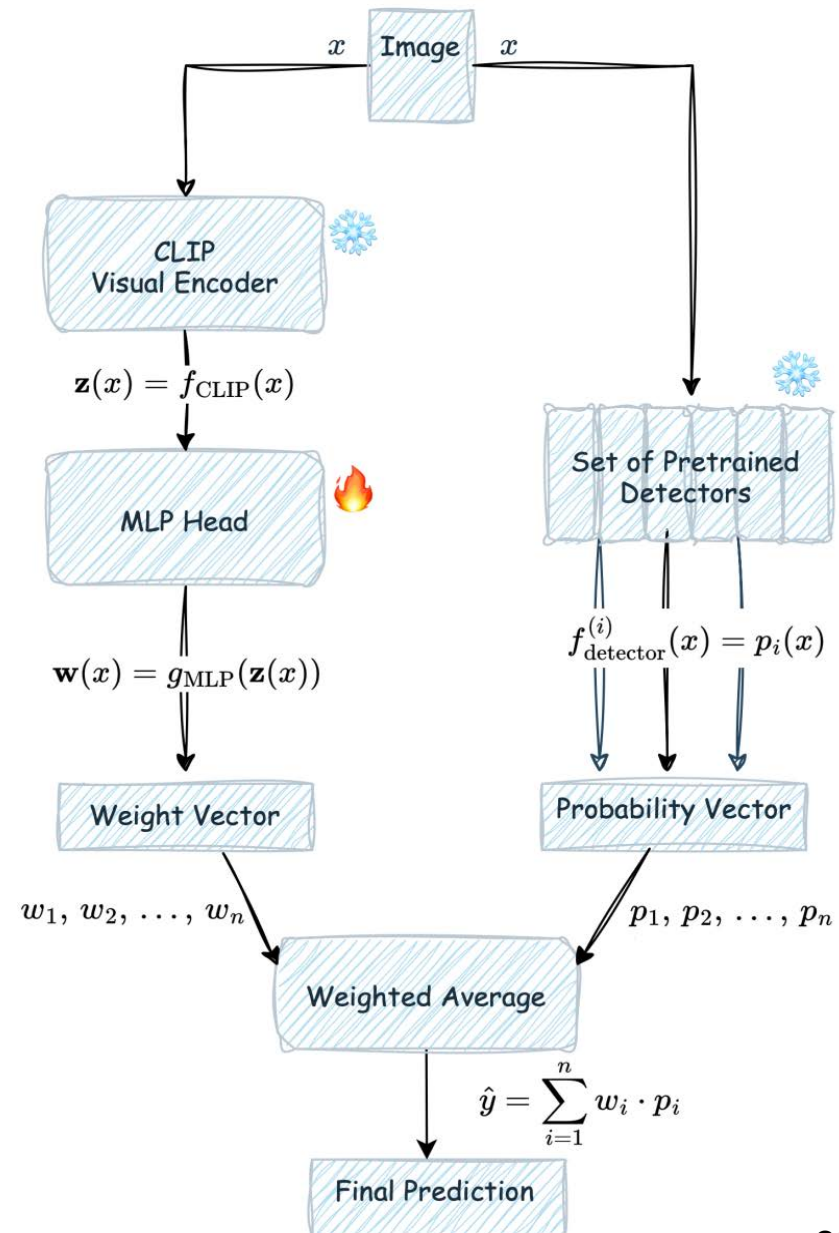
Fake



Ensemble of Deepfake Detectors



- Many deepfake detectors available, but their generalization unclear
- Ensemble model
 - Combines multiple detector
 - The combination is learned
- Model architecture
 - CLIP image encoder
 - Dynamic weights predict by the model
 - Final prediction: weighted average of detector softmax probabilities (convex combination)
 - [\[Petrilakova-2026\]](#)



Ensemble of Deepfake Detectors



■ Experiments

Test datasets

Generator	Year	Source
Fake images		
GAN-based generators		
StyleGAN3	2021	DF40 dataset
FF++ (manipulations)	2019	DeepfakeBench
WhichFaceIsReal	2019	DF40 dataset
Diffusion-based generators		
PixArt-alpha	2023	Generated
Realistic Vision v6	2024	Generated
Stable Diffusion XL	2023	Generated
Kandinsky 3.0	2023	Generated
FLUX-dev	2024	Generated
DeepFloyd IF	2023	Generated
Commercial generators		
MidJourney v6	2023	DF40 dataset
Real images		
Real mix	2015-19	IMDB+CelebA+FFHQ

Detectors (13 pre-trained DF detectors)

Detector Name	Year	Publication
CNND, CNND2	2020	CVPR
FreqD	2020	ICML
GramNet	2020	CVPR
Fusing	2022	ICIP
DIMD-latent, DIMD-gan	2023	ICASSP
UnivFakeD	2023	CVPR
RPTC	2023	ArXiv
DeFake	2023	SIGSAC
NPR	2024	CVPR
ClipBased, Corvi2023	2024	CVPRw
DeepfakeD	2026	WACV

Ensemble of Deepfake Detectors



Results

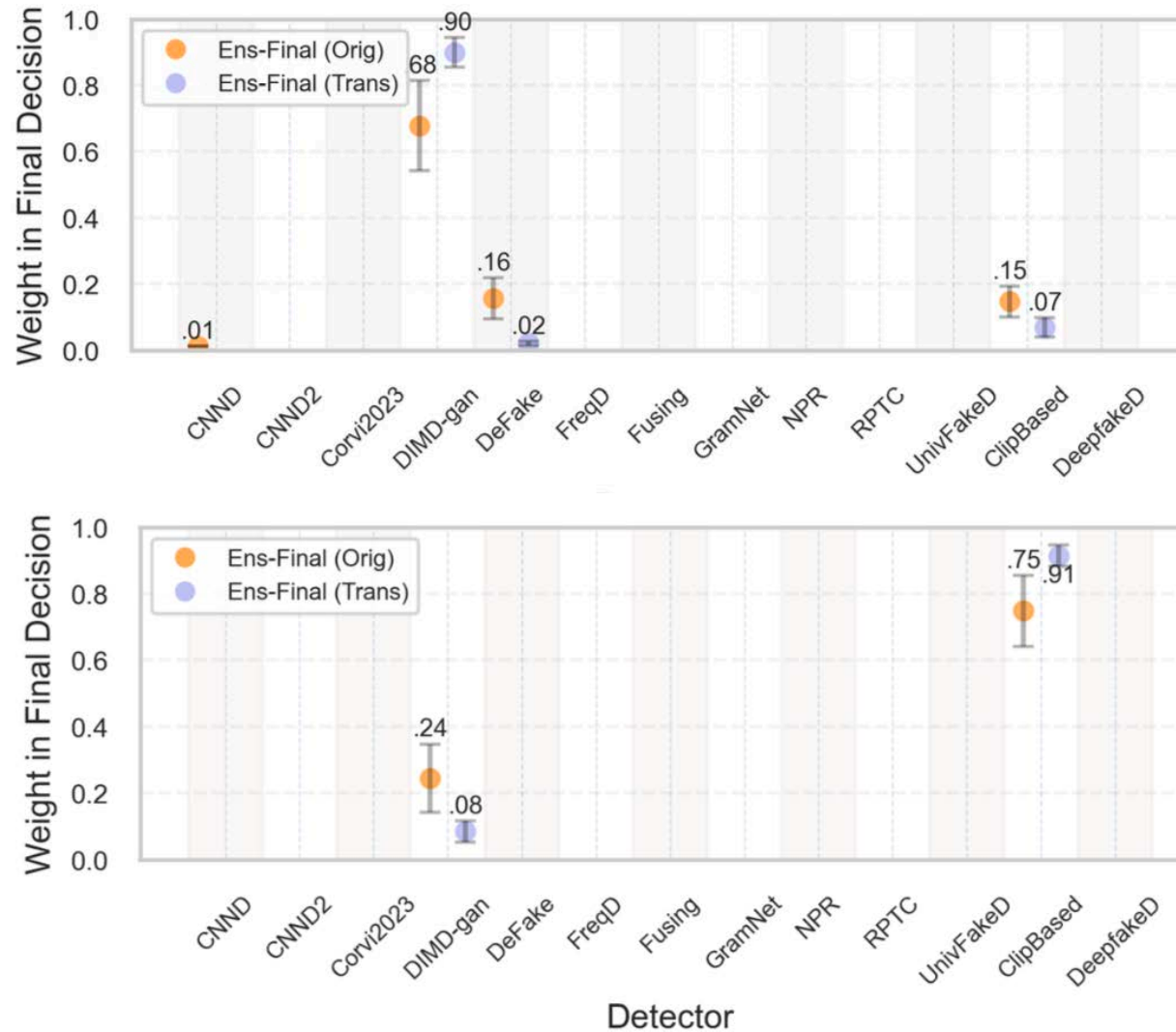
Detector	DF-IF	MidJ	RV	SD-XL	Flux	Kndsky	PixArt	FF++	StyleG3	WFIR	Row Avg
CNND	0.51	0.52	0.50	0.50	0.50	0.50	0.51	0.50	0.52	0.51	0.51
CNND2	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50
Corvi2023	0.57	0.67	0.61	0.64	0.58	0.70	0.56	0.59	0.80	0.66	0.64
DIMD-gan	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.51	0.50	0.50
DIMD-latent	0.51	0.63	0.64	0.59	0.53	0.62	0.53	0.51	0.58	0.58	0.57
DeFake	0.59	0.63	0.65	0.67	0.48	0.73	0.67	0.52	0.41	0.61	0.60
FreqD	0.50	0.61	0.46	0.55	0.49	0.59	0.62	0.46	0.46	0.54	0.53
Fusing	0.54	0.54	0.50	0.52	0.53	0.50	0.53	0.50	0.62	0.52	0.53
GramNet	0.53	0.51	0.52	0.52	0.51	0.52	0.52	0.47	0.47	0.51	0.51
NPR	0.50	0.50	0.50	0.50	0.50	0.51	0.50	0.50	0.50	0.50	0.50
RPTC	0.60	0.58	0.42	0.58	0.61	0.47	0.49	0.55	0.54	0.58	0.54
UnivFakeD	0.48	0.48	0.48	0.48	0.48	0.52	0.49	0.61	0.62	0.49	0.51
ClipBased	0.58	0.54	0.58	0.58	0.56	0.59	0.58	0.59	0.59	0.54	0.57
DeepfakeD	0.54	0.59	0.47	0.55	0.52	0.64	0.64	0.89	0.93	0.47	0.62
Ens-Small	0.61	0.64	0.75	0.68	0.58	0.79	0.66	0.86	0.88	0.55	0.67
Ens-Probs	0.57	0.57	0.60	0.57	0.55	0.59	0.56	0.50	0.51	0.56	0.56
MonoCLIP	0.54	0.55	0.81	0.57	0.55	0.58	0.55	0.79	0.73	0.51	0.57
Ens-Final	0.68	0.66	0.89	0.76	0.69	0.87	0.85	0.98	0.95	0.54	0.82

=> Ensemble model outperforms all individual detectors



Ensemble of Deepfake Detectors

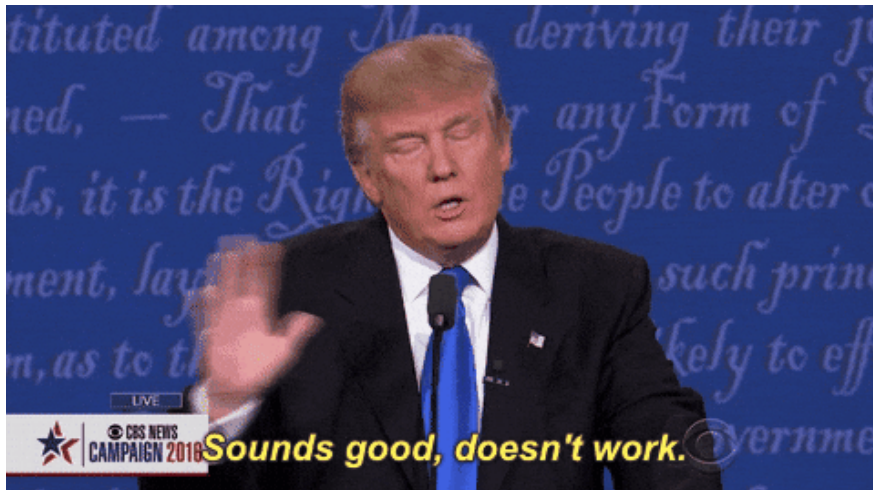
- Activated weights





Deepfake Detection by Motion Anomalies

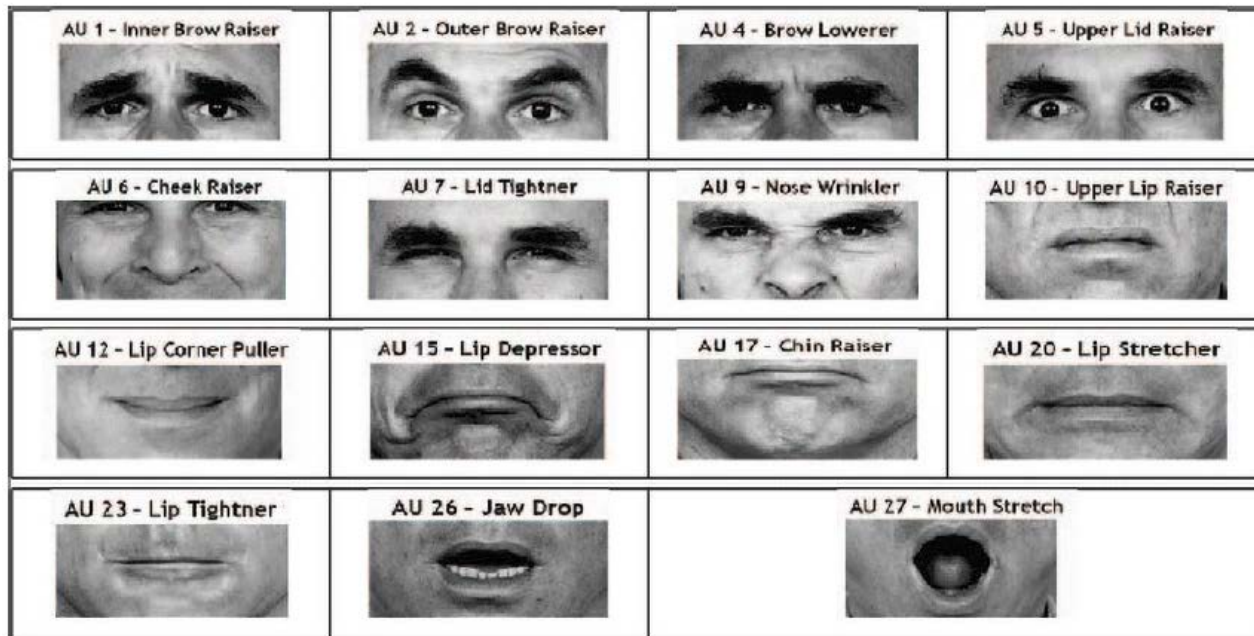
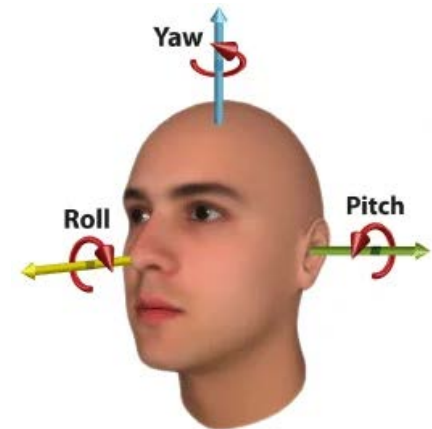
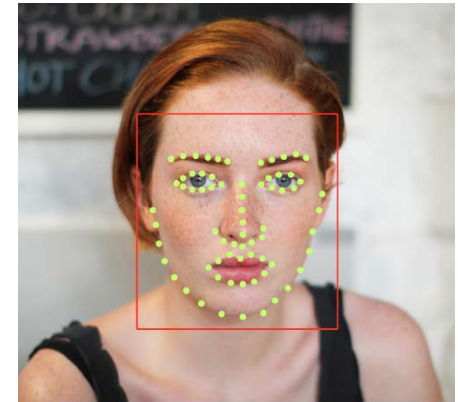
- **Idea: “Motion is not faked perfectly”**
 - Detecting imperfect motion is a strong cue
 - Inconsistent motion across frames
 - Lips, eye-blinks, eyebrows, etc. not synced with the speech
 - Unnatural motion patterns (**person specific models**)
 - Each person performs a characteristic motion when speaking – Talking Motion Signatures





Talking Motion Signatures (TMS)

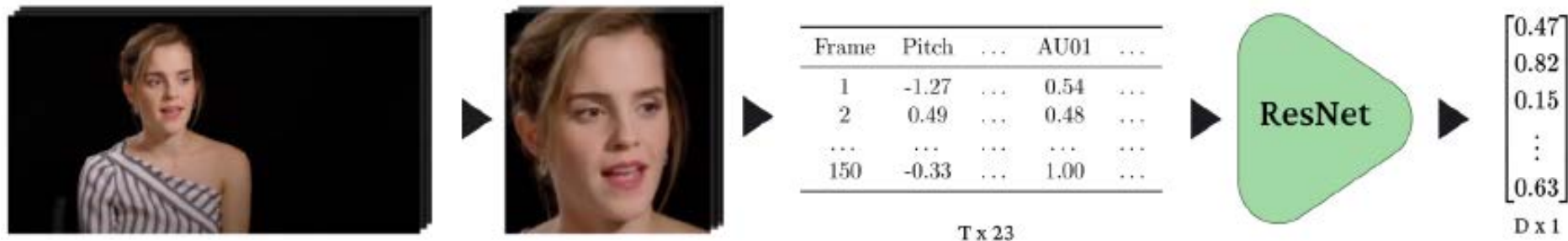
- Paper [[Samarskyi-2026](#)]
- Can we identify individuals by talking motions only? (without appearance)
- Facial Action Units (FAUs) [[Ekman-1978](#)]
 - Facial muscle activations
- Head orientation (roll, pitch, yaw)





Talking Motion Signatures (TMS)

- Dataset of individuals speaking
 1. TalkingCelebs: 5 subjects (100 videos/identity, ~30 secs.)
 2. VoxCeleb2: 630 subjects (12 videos/identity, ~ 20 secs.)
- TMS baseline – All temporal cross-correlations [[Bohacek-2022](#)]
- TMS proposed – Learned deep net (ResNET-50) ~penultimate layer

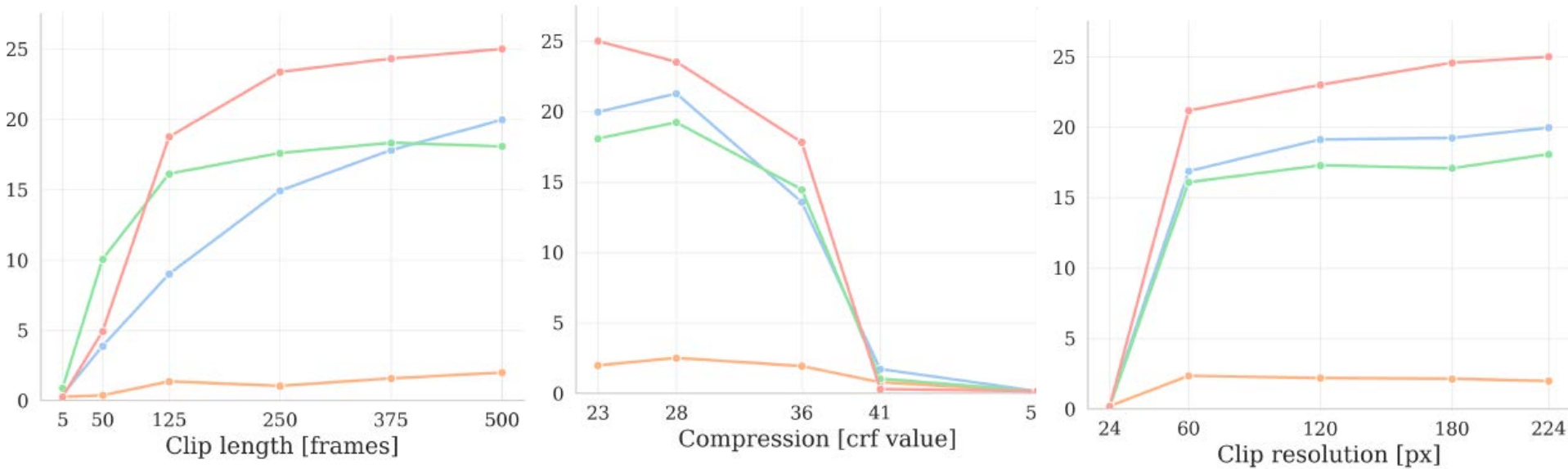


Talking Motion Signatures (TMS)



■ ID-Recognition Accuracy (VoxCeleb2)

—●— SVM —●— One-class SVM —●— ResNet50 —●— ArcFace ResNet50



=> Works for short videoclips (~5 seconds)

=> Robust to adverse condition (until landmarks/FAU extractor works)



TMS: Impersonation Quality

- Impersonation quality assessment
- Target identity has a prototype learned by the model (Barack Obama)
- Cosine similarity with the prototype is computed with other actor's TMS extracted from the query video

$$\frac{1}{N} \sum_{t=1}^N \mathbf{p}^T \mathbf{q}(t)$$



Dwayne Johnson
Similarity: -0.28



Jordan Peele
Similarity: 0.87



Reggie Brown
Similarity: 0.87



Steve Bridges
Similarity: 0.42

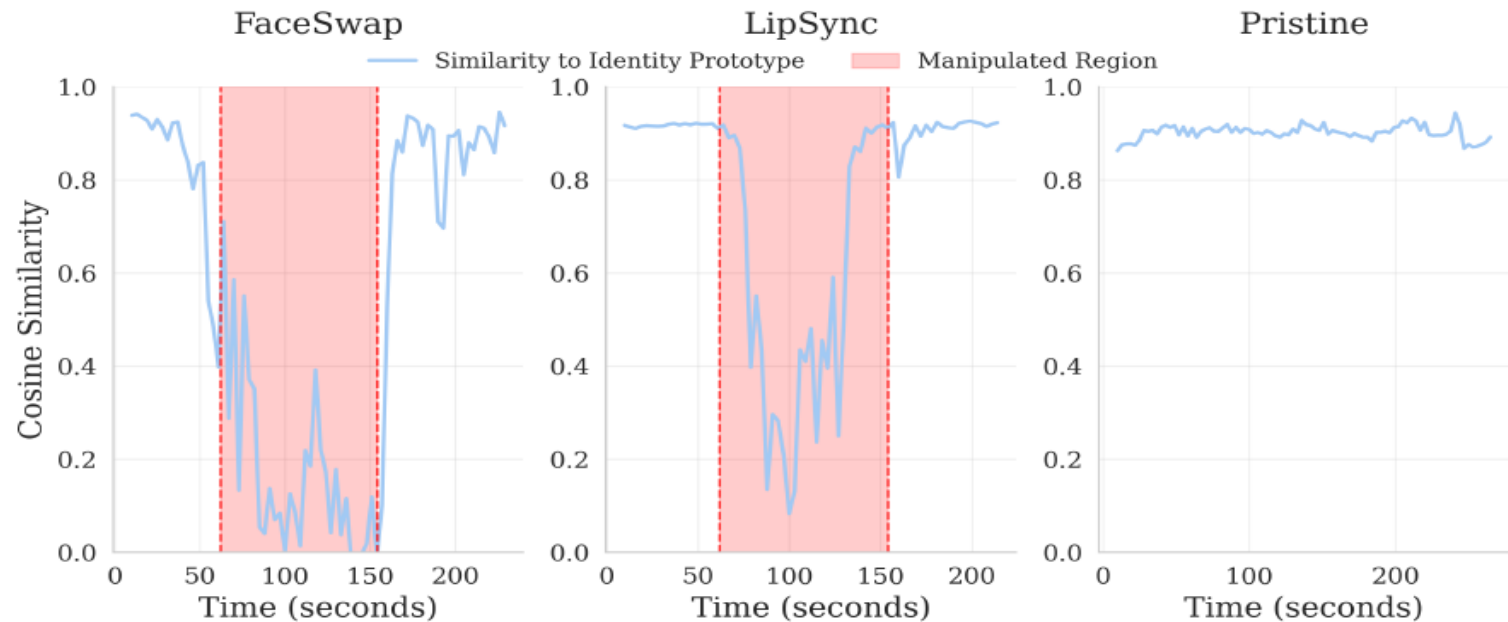
- Interestingly, the cosine similarity correlates with the number of followers on YouTube



TMS: Deepfake Detection

- Target identity has a prototype learned by the model (V. Zelenskyj)
- Cosine similarity (prototype-query) calculated over time

$$\mathbf{p}^T \mathbf{q}(t)$$



- ⇒ TMS-DF detector uses high-level semantics
- Does not overfit to a particular generator (generalization)

Conclusions



- Multiple easy-to-use tools to produce deepfakes in a large scale
- Deepfake detection – good news
 - Easy detection/localization for in-distribution data (seen generators)
- Deepfake detection challenges – bad news
 - Generalization to unseen detectors is a major problem (cross-dataset)
 - Compression/Blur/Subsampling tend to erase the generator fingerprint
 - Detector is very sensitive to adversarial attacks
- Simple architecture achieving SotA accuracy (GenD)
- Ensemble outperforms individual detectors
- Promising results of “Talking motion signature” DF detector
 - Complimentary approach relying on higher-level features of the video